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A METHOD FOR EXPERIMENTAL CORRELATION DISTRIBUTION ESTIMATION FOR CORRELATION-BASED DIGITAL IMAGE WATERMARKS

V. Anzhin[1]

[1] *Tomsk State University, Tomsk, Russian Federation*

This paper proposes a method for experimentally estimating the distribution of correlation in correlation-based digital image watermarking. The proposed method evaluates how distortions affect the correlations between image and embedded and non-embedded watermarks. The method involves a computational experiment using an image dataset, where correlations are measured before and after filtering for embedded and non-embedded watermarks. Applying the proposed method to a DCT based watermarking algorithm demonstrates that filtering significantly reduces the correlation with the embedded watermark, while having minimal changes for non-embedded. **Keywords:** *watermark, robustness, correlation measure*

Introduction

The growth in digital multimedia production and consumption has increased the demand for technical solutions to protect copyrights and restrict unauthorized copying. Digital watermarks can be used for multimedia content protection. In the case of a content leak, digital watermarks allow to verify authorship and restrict the distribution of pirated copies.

In correlation-based watermarking [1], a pseudo-random signal is embedded into the image as watermark, and a correlation measure is used for watermark detection. The image is identified as watermarked when the correlation between it and the embedded watermark exceeds the selected detection threshold.

The image with embedded watermark can be modified by various distortions. These distortions can result from transmission losses and signal processing. They can also arise from attempts to remove the watermark. Watermark robustness [2] refers to the resilience of the watermark against various signal processing operations. Distortion of the image changes the value of correlation between the watermark and the distorted image. This can lead to the omission of embedded watermark (false-negative detection) or the false detection of non-embedded watermark (false-positive detection).

An important aspect of watermark robustness is the right detection threshold selection. Selecting an appropriate detection threshold requires an understanding of the distribution of correlation between the watermark and both watermarked and non-watermarked images. There are theoretical models which use statistical methods, typically developed for a specific watermarking method. For example, in [3], the correlation is estimated and detection threshold is selected through theoretical analysis for embedding method which uses discrete cosine transform (DCT). The authors of [3] also note that the selected threshold can lead to a significant number of false-negative detections under attack. Estimating the detection threshold theoretically is a complex task due to the wide variety of potential distortions that can affect the image. As an alternative, computational experiments can be used for modelling correlation distribution and detection threshold selection.

This paper proposes the method for experimentally estimating the distribution of correlation in correlation-based image watermarking algorithms. The method involves watermark embedding, applying filtering to introduce distortions, and calculating the correlation measure for both watermarked and non-watermarked images before and after filtering. The proposed method is applied to the watermarking algorithms described in [4].

The C++ implementation of the watermarking algorithm, along with the Python scripts of experiments and the obtained results, are available at [5].

1. DCT Based Watermarking

Within the DCT based watermarking algorithm [4] the original image is divided into blocks of 8x8 pixels. A DCT is applied to each block. The transformed block is then converted into a sequence of 64 elements using ZigZag scanning [6], which reorders the DCT coefficients from low to high frequencies. In each sequence, the most significant low-frequency DCT coefficients are discarded. The remaining DCT coefficients are concatenated into a single sequence c_o. The length of c_o is n. The watermark w_r is embedded into the sequence c_o by the equation (1). The watermark w_r elements are generated according to the normal distribution $N(0, 1)$. The modified blocks are transformed via inverse DCT to obtain the watermarked image.

$$c_w = c_o + \alpha w_r, \tag{1}$$

where $c_w, c_o, w_r \in R^n$, $n \in N$, $\alpha \in R$.

To detect the watermark w_r the normalized correlation (3) is calculated between w_r and the sequence of DCT coefficients c extracted from image.

$$\tilde{c}[i] = \frac{c[i]}{\sqrt{\sum\limits_{j=1}^{n} c[j]^2}}, \qquad \tilde{w}_r[i] = \frac{w_r[i]}{\sqrt{\sum\limits_{j=1}^{n} w_r[j]^2}}, \tag{2}$$

$$Z_{nc}(c, w_r) = \sum_{i=1}^{n} \tilde{c}[i]\tilde{w}_r[i], \tag{3}$$

where $c, w_r, \tilde{c}, \tilde{w}_r \in R^n$, $n \in N$.

2. Data collection method of evaluating the distribution of correlation

A computational experiment on a set of images is conducted to evaluate the distribution of correlation. Two random watermarks are generated for each image in the set. One watermark is embedded into the image. The correlation measures are calculated between the watermarked image and both the embedded and non-embedded watermarks. The watermarked image is processed with a filter. Correlations are calculated between the filtered image and both the embedded and non-embedded watermarks. The output of the algorithm is the arrays of correlation values. The method is described in Algorithm 1.

Algorithm 1. Data collection method

Require: Watermark embedding algorithm $W(c_o, w_r)$, correlation measure $Z(c, w_r)$, set of image filters F, images database I

Ensure: Z_e – array of correlations between the watermarked image and the embedded watermark, Z_{ne} – array of calculated between the image with embedded watermark and a different non-embedded watermark, Z_{ef} and Z_{nef} – arrays of correlations after filtering with $f \in F$

$j \leftarrow 0$
for all $i \in I$
  Create random watermarks w and w' for image i, $w \neq w'$
  $i_w = W(i, w)$
  $Z_e[j] \leftarrow Z(i_w, w)$, $Z_{ne}[j] \leftarrow Z(i_w, w')$
  for all $f \in F$
    $i_{wf} = f(i_w)$
    $Z_{ef}[j] \leftarrow Z(i_{wf}, w)$, $Z_{nef}[j] \leftarrow Z(i_{wf}, w')$
  $j \leftarrow j + 1$

3. Testing results

The computational experiment, described in Algorithm 1 was conducted using the watermarking algorithm [4]. The Flickr8K dataset [7] with 8091 images was used. The following FFmpeg filters [8] were used to filter the images:

- Blurring using `unsharp`: `unsharp=3:3:-0.25:3:3:-0.25`;
- Sharping using `unsharp`: `unsharp=3:3:0.25:3:3:0.25`;
- DCT based denoising using `dctdnoiz`: `dctdnoiz=s=10`;
- Denoising using `nlmeans`: `nlmeans=s=10`.
- Pixelation using `pixelize`: `pixelize=w=3:h=3`;
- Denoising using `fftdnoiz`: `fftdnoiz=sigma=15`.

The Table 1 presents the mean and standard deviation values calculated for the array of correlation values generated by Algorithm 1. The first column indicates the α values used during watermark embedding. The second column specifies the filter applied to the images. Columns under the heading "With Embedding" contains the mean and standard deviation of correlation values between an image and an embedded watermark. The columns under "Without Embedding" show the results between an image and a non-embedded watermark.

The results of Algorithm 1 can be presented in a chart form. Fig. 1 shows the estimated distribution of correlation calculated using the Flickr8K dataset [7] with embedding performed at $\alpha = 10$. The chart on the left displays the correlation values without filtering, and the chart on the right shows the correlation values after filtering with dctdnoiz. The x-axis represents the correlation values, and the y-axis is the number of images with the corresponding correlation value. Orange color is used for cases where the detected watermark was not embedded into the image, and blue is used when the watermark was embedded. Dashed lines shows the mean values.

The results in Table 1 show that when a watermark is embedded into an image, the mean of correlation values is higher than the mean for a non-embedded watermark. Applying filters to the images generally decreases the mean of correlation values for the embedded watermark (with the exception of the sharpening filter), while producing minimal changes in the mean values for the non-embedded watermark. In the chart in Fig. 1, the orange distribution (correlation values for the non-embedded watermark) remains close to zero after filtering, whereas the blue distribution (correlation values for the embedded watermark) shifts toward lower correlations.

Table 1

Correlation estimation results

α	Filter	With Embedding		Wihtout Embedding	
		Mean	Std dev	Mean	Std dev
5	without filtering	9.84×10^{-7}	3.82×10^{-7}	2.25×10^{-11}	4.30×10^{-9}
	blur	9.32×10^{-7}	3.60×10^{-7}	2.73×10^{-11}	4.30×10^{-9}
	sharp	1.01×10^{-6}	3.93×10^{-7}	2.14×10^{-11}	4.31×10^{-9}
	dctdnoiz	1.14×10^{-7}	4.28×10^{-8}	3.64×10^{-11}	4.26×10^{-9}
	nlmeans	2.11×10^{-7}	8.83×10^{-8}	4.07×10^{-11}	4.31×10^{-9}
	pixelize	5.06×10^{-8}	1.62×10^{-8}	-4.22×10^{-12}	4.34×10^{-9}
	fftdnoiz	3.31×10^{-7}	1.09×10^{-7}	1.72×10^{-11}	4.25×10^{-9}
10	without filtering	1.55×10^{-6}	4.49×10^{-7}	-6.10×10^{-12}	4.35×10^{-9}
	blur	1.50×10^{-6}	4.40×10^{-7}	-1.08×10^{-11}	4.34×10^{-9}
	sharp	1.57×10^{-6}	4.53×10^{-7}	-7.70×10^{-12}	4.35×10^{-9}
	dctdnoiz	2.99×10^{-7}	9.59×10^{-8}	-2.35×10^{-13}	4.33×10^{-9}
	nlmeans	4.70×10^{-7}	1.66×10^{-7}	8.29×10^{-12}	4.30×10^{-9}
	pixelize	1.00×10^{-7}	3.17×10^{-8}	9.54×10^{-12}	4.38×10^{-9}
	fftdnoiz	1.02×10^{-6}	2.97×10^{-7}	-1.64×10^{-11}	4.31×10^{-9}
15	without filtering	1.87×10^{-6}	4.35×10^{-7}	-6.63×10^{-12}	4.33×10^{-9}
	blur	1.83×10^{-6}	4.32×10^{-7}	-4.53×10^{-12}	4.33×10^{-9}
	sharp	1.88×10^{-6}	4.36×10^{-7}	-8.24×10^{-12}	4.32×10^{-9}
	dctdnoiz	9.35×10^{-7}	3.09×10^{-7}	-5.13×10^{-11}	4.34×10^{-9}
	nlmeans	8.73×10^{-7}	2.41×10^{-7}	-4.07×10^{-11}	4.37×10^{-9}
	pixelize	1.50×10^{-7}	4.76×10^{-8}	-4.05×10^{-11}	4.29×10^{-9}
	fftdnoiz	1.60×10^{-6}	4.02×10^{-7}	-2.34×10^{-11}	4.33×10^{-9}

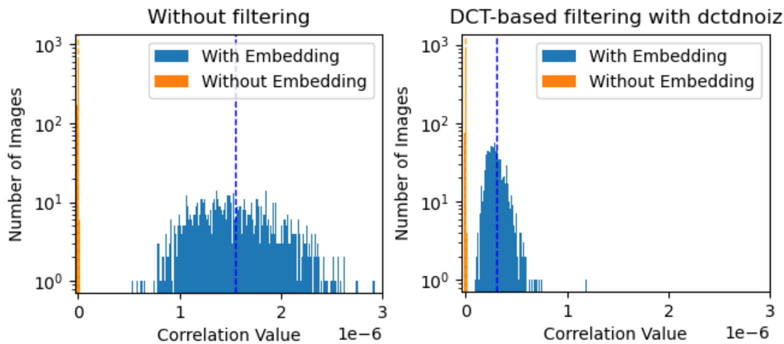

Figure 1. Chart of correlation without filtering (left) and after filtering with dctdnoiz (right)

4. Conclusion

This paper proposed a method for the experimental estimation of the correlation distribution in correlation-based image watermarking. The proposed approach enables analysis of how various distortions affect the correlation between watermarked image and embedded and non-embedded watermarks. The method was applied to a DCT based watermarking algorithm [4] using the Flickr8K dataset [7] and a set of FFmpeg filters [8]. The implementation and experimental results are available at [5].

The results show that distortions introduced by filters significantly reduces the correlation values between the embedded watermark and the distorted watermarked image, while having minimal changes in the correlation between the distorted image and non-embedded watermarks.

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Viktor Anzhin — Ph.D. student, Institute of Applied Mathematics and Computer Science, TSU. E-mail: *viktor.anjin@gmail.com*